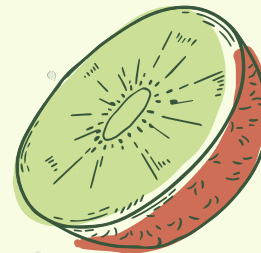
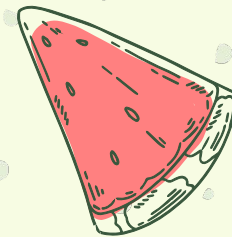


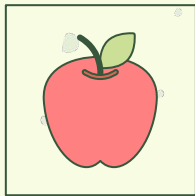
FEEDING AMERICA: MAP THE MEAL GAP

Presented by Jake Mecca, Sarah Toniuk



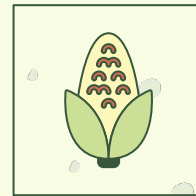


BEHIND THE DATA



What is the data?

- Annual county and state-level estimates of food insecurity across the U.S. (2019-2023)
- Includes key indicators related to food insecurity, such as:
 - SNAP eligibility thresholds
 - Average cost per meal
 - Food insecurity rates by age, race, and ethnicity



What is the origin of the data?

- Combines Census survey data with economic and demographic indicators.
 - Uses statistical modeling to estimate food insecurity rates for each state/county.
 - Dataset produced by Feeding America in collaboration with researchers and public data sources.
- 

WHY DID WE CHOOSE **THIS DATASET?**



It's important

Food is a basic necessity.
Nobody should be
insecure about access.



Recent Related Events

With the Big Beautiful Bill going into effect and SNAP restrictions, now is the time to make this a conversation.



It's a timeless issue

The organization of origin
has data from all the way
back to 2009.



There's a lot to work with

The dataset has many columns
to draw conclusions and
correlations from.



DATASET OVERVIEW

Overall Food Insecurity Rate

Percentage of inhabitants who are food insecure

of Food Insecure Persons

Number of inhabitants who are food insecure

Food Insecurity Among...

The percentage of inhabitants who are food insecure, separated by: Black, Hispanic, and white/non-Hispanic

Cost Per Meal

The average dollar amount spent on food per meal by food-secure individuals

SNAP Threshold %

Maximum gross income that households can have to qualify, as a percent of the federal poverty guidelines for SNAP

State	Year	Overall Food Insecurity Rate	# of Food Insecure Persons	Food Insecure Persons by Race	SNAP Threshold %	Food Insecurity Among...	Cost Per Meal												
AL	2019	10.10%	78,250	27%	13%	12%	10%	50%	50%	21.20%	229,880	76%	24%	9.40%	105,776	15.60%	105,533	\$3.01	
AL	2020	14.50%	71,370	27%	14%	10%	10%	53%	48%	19.80%	216,040	82%	18%	9.00%	104,269	14.20%	97,677	\$3.13	
AL	2021	14.80%	74,550	26%	12%	10%	10%	41%	60%	18.30%	204,830	77%	23%	8.90%	103,567	14.40%	100,289	\$3.50	
AL	2022	17.40%	88,270	31%	18%	12%	10%	39%	61%	23.30%	258,560	75%	25%	9.20%	109,300	13.40%	87,500	\$3.91	
AL	2023	17.50%	89,510	31%	19%	13%	10%	48%	52%	22.20%	250,580	75%	25%	8.10%	103,465	10.80%	62,405	\$3.50	
AK	2019	11.90%	8,970	18%	14%	9%	10%	49%	51%	16.10%	28,850	72%	28%	6.70%	8,609	10.70%	9,576	\$3.63	
AK	2020	10.80%	7,670	20%	15%	7%	10%	50%	50%	14.70%	26,910	77%	23%	5.90%	7,863	11.70%	10,052	\$3.81	
AK	2021	10.70%	7,260	19%	12%	8%	10%	40%	60%	12.80%	22,960	73%	27%	8.40%	12,149	8.00%	6,184	\$4.25	
AK	2022	12.80%	9,620	23%	19%	9%	10%	37%	63%	17.10%	30,110	71%	29%	7.50%	11,900	7.00%	4,800	\$4.70	
AK	2023	14.50%	10,080	26%	21%	10%	10%	45%	55%	17.80%	31,030	67%	33%	7.50%	11,566	6.30%	4,674	\$4.23	
AZ	2019	12.60%	91,940	22%	17%	10%	15%	63%	37%	17.90%	293,570	68%	32%	8.50%	135,977	10.40%	91,931	\$3.03	
AZ	2020	11.10%	79,790	22%	17%	8%	15%	67%	33%	16.40%	269,610	76%	24%	6.90%	118,018	10.00%	85,230	\$3.07	
AZ	2021	10.30%	75,690	20%	14%	8%	15%	49%	51%	14.00%	226,080	68%	32%	7.40%	129,412	8.50%	79,721	\$3.30	
AZ	2022	13.10%	96,130	24%	19%	11%	15%	47%	53%	18.50%	293,950	65%	35%	9.40%	159,300	7.20%	71,500	\$3.68	
AZ	2023	14.40%	1,06,120	27%	21%	11%	15%	58%	42%	18.40%	291,290	64%	36%	10.50%	182,986	11.60%	106,514	\$3.32	
AR	2019	16.60%	49,950	25%	19%	15%	10%	51%	49%	21.60%	150,910	73%	27%	8.20%	57,508	12.90%	47,370	\$2.73	
AR	2020	14.70%	44,130	25%	19%	13%	10%	53%	47%	19.70%	138,410	77%	23%	6.30%	45,417	16.90%	61,190	\$2.80	
AR	2021	15.50%	46,550	25%	16%	13%	10%	41%	59%	19.10%	134,690	73%	27%	7.60%	55,600	19.80%	75,302	\$3.18	
AR	2022	18.60%	56,110	30%	22%	16%	130%	39%	61%	24.20%	168,430	71%	29%	11.30%	83,400	18.60%	73,700	\$3.55	
AR	2023	19.30%	59,860	33%	23%	17%	130%	45%	55%	24.00%	167,940	73%	27%	13.20%	101,897	17.00%	60,239	\$3.22	
CA	2019	10.20%	4,01,960	19%	15%	8%	200%	72%	28%	13.70%	1,221,560	70%	30%	6.30%	506,182	9.90%	478,042	\$3.26	
CA	2020	9.10%	3,57,920	19%	16%	6%	200%	77%	23%	13.00%	1,165,400	79%	21%	6.50%	541,877	10.00%	475,378	\$3.45	
CA	2021	10.50%	4,10,060	18%	15%	10%	200%	78%	22%	13.00%	1,165,400	79%	21%	6.50%	541,877	10.00%	475,378	\$3.45	
CA	2022	12.60%	4,91,450	23%	17%	13%	200%	78%	22%	13.00%	1,165,400	79%	21%	6.50%	541,877	10.00%	475,378	\$3.45	
CA	2023	13.70%	5,35,790	25%	18%	14%	200%	78%	22%	13.00%	1,165,400	79%	21%	6.50%	541,877	10.00%	475,378	\$3.45	
CO	2019	9.80%	56,440	12%	8.30%	47,420	13%	9.20%	533,650	14%	11.50%	671,340	19%	12.70%	744,950	21%	12.00%	428,800	22%
CO	2020	10.20%	364,040	23%	10.50%	380,310	21%	12.90%	468,150	25%									

ons

e

Food Insecurity Among...

The percentage of inhabitants



HOW WE **CLEANED** OUR DATA

■ **Renamed Columns for Ease**

Made everything easier to work with when coding

■ **Mutated Values**

Dropped any “%” “,” “\$” etc (parsed numbers for calculations)

■ **Added Racial Gap Columns**

The racial gap columns were created by calculating the difference between White and Black food insecurity rates, as well as the difference between White and Hispanic food insecurity rates.

```
food <- food %>%
  rename(
    fi_rate_overall = 'Overall Food Insecurity Rate',
    fi_num_overall = '# of Food Insecure Persons Overall',
    black-fi_rate = 'Food Insecurity Rate among Black Persons (all ethnicities)',
    hispanic-fi_rate = 'Food Insecurity Rate among Hispanic Persons (any race)',
    nonhispanic-fi_rate = 'Food Insecurity Rate among White, non-Hispanic Persons',
    snap_threshold = 'SNAP Threshold in State',
    pct-fi_below_snap = '% FI ≤ SNAP Threshold',
    pct-fi_above_snap = '% FI > SNAP Threshold',
    child-fi_rate = 'Child Food Insecurity Rate',
    fi_children = '# of Food Insecure Children',
    pct-fi_child_below_185fpl = '% food insecure Children in HH w/HH Incomes Below 185 FPL',
    pct-fi_child_above_185fpl = '% food insecure Children in HH w/HH Incomes Above 185 FPL',
    fi_rate_senior = 'Senior Food Insecurity Rate',
    fi_seniors = '# of Food Insecure Seniors',
    fi_rate_older_adult = 'Older Adult Food Insecurity Rate',
    fi_older_adults = '# of Food Insecure Older Adults',
    cost_per_meal = 'Cost Per Meal',
    weekly_budget = 'Weighted weekly $ needed by FI',
    annual_budget_shortfall = 'Weighted Annual Food Budget Shortfall',
    census_division = 'Census Division',
    census_region = 'Census Region',
    fns_region = 'FNS Region'
  ) %>% mutate(across(c(fi_rate_overall,
    black-fi_rate,
    hispanic-fi_rate,
    nonhispanic-fi_rate,
    child-fi_rate,
    fi_rate_senior,
    fi_rate_older_adult,
    pct-fi_below_snap,
    pct-fi_above_snap,
    pct-fi_child_below_185fpl,
    pct-fi_child_above_185fpl), ~ parse_number(.x)))

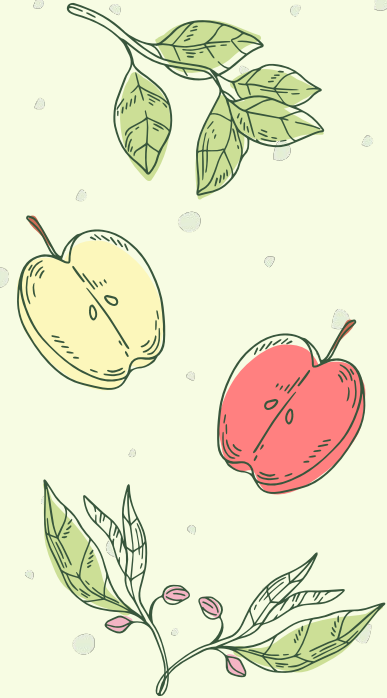
food <- food %>%
  filter(!is.na(black-fi_rate), !is.na(hispanic-fi_rate), !is.na(nonhispanic-fi_rate)) %>%
  mutate(
    gap_black_white = black-fi_rate - nonhispanic-fi_rate,
    gap_hisp_white = hispanic-fi_rate - nonhispanic-fi_rate
  )
```



WHAT FACTORS CONTRIBUTE MOST TO RACIAL DISPARITIES IN FOOD INSECURITY?

We used Linear Regression Modeling and made Visualizations to understand and answer the following:

1. Is it just that states with high overall FI have bigger gaps?
2. Do economic factors (meal cost) matter more than policy factors (SNAP)?
3. Are the main reasons for the Black-White gap the same as the Hispanic-White gap?

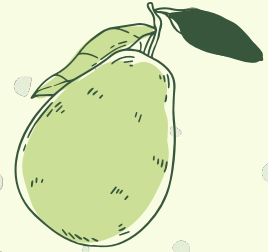




NUTRITION



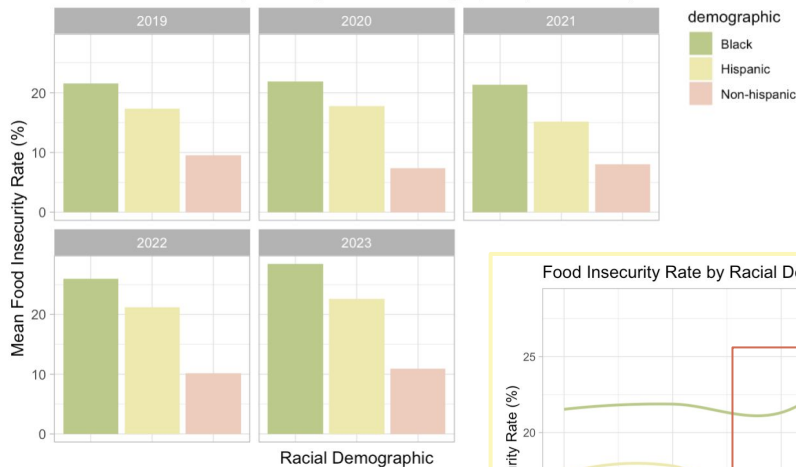
DATA VIZ!



FOOD

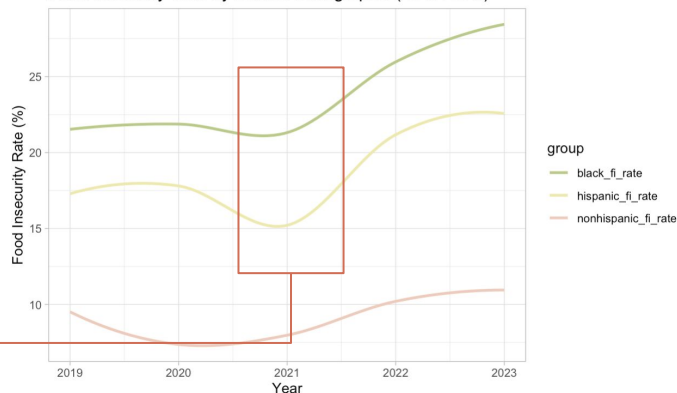
FOOD INSECURITY RATE BY RACIAL DEMOGRAPHIC

Mean Food Insecurity Rate by Racial Demographic (2019-2023)



Increase in government assistance programs in response to the COVID-19 pandemic

Food Insecurity Rate by Racial Demographic (2019-2023)

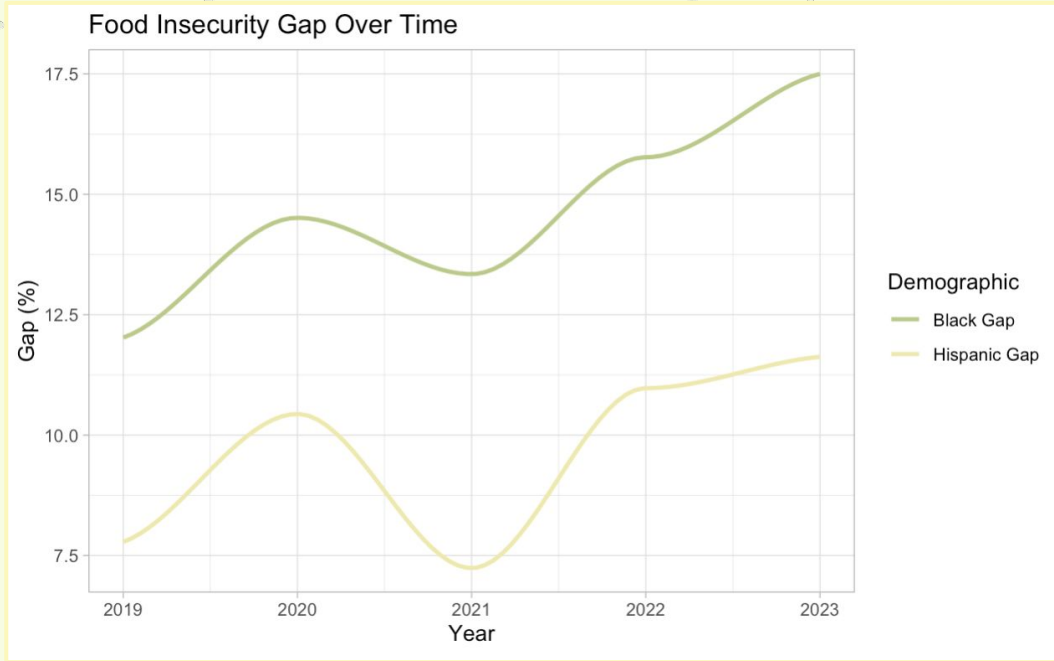


- Clear disparities exist in food insecurity rates for Black and Hispanic populations.
- Black food insecurity rates are the highest each year, reaching an all-time high of about 28% in 2023.

- Food insecurity rates have risen over time, but at noticeably different rates across racial groups.
- Both Black and Hispanic food insecurity rates show a noticeable dip in 2021.



RACIAL/ETHNIC GAPS OVER TIME



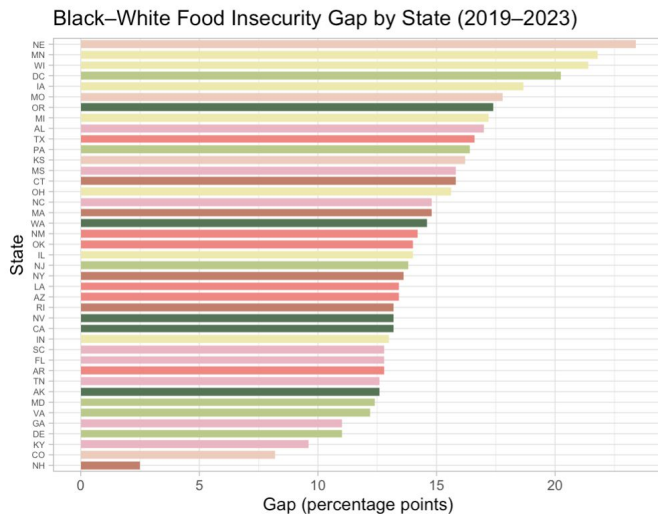
Using the food insecurity gap variables we created, this graph provides a clearer visualization on how much the racial gap in food insecurity as increased over time.



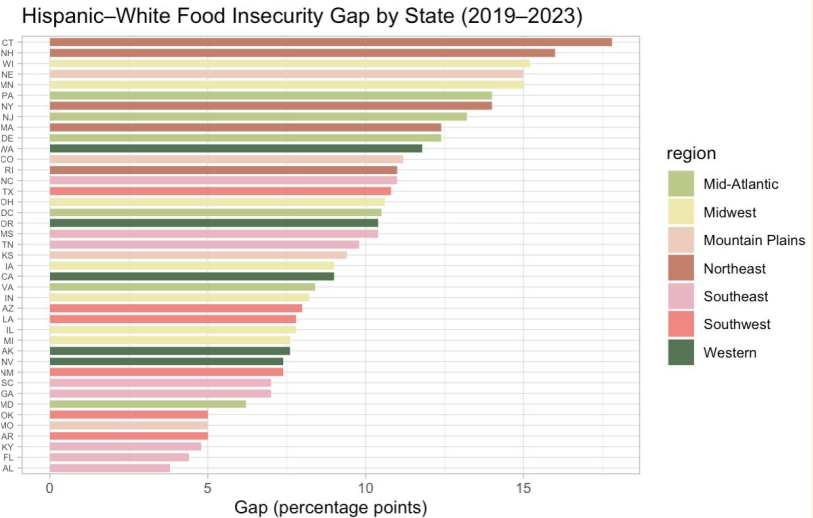


RACIAL/ETHNIC GAPS BY REGION

#1: Nebraska



#1: Connecticut



region

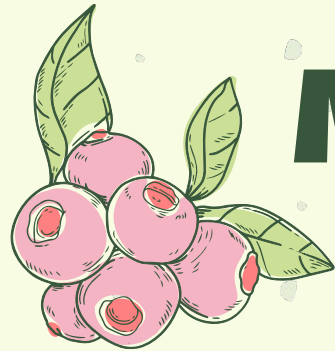
- Mid-Atlantic
- Midwest
- Mountain Plains
- Northeast
- Southeast
- Southwest
- Western

- Minnesota, Nebraska, and Wisconsin all show elevated gaps for both Black and Hispanic food insecurity rates.
- No clear regional conclusions that could be made from the data, especially due to the lack of population data.





NUTRITION



MODELING!



FOOD



REGRESSION // BOOTSTRAP

**Regression modeling was controlled on region, as census (midwest v northeast v south v west) regional differences played a big role and made other data seem insignificant.

Black/
White

R^2 :
0.4217
Adjusted:
0.4041

Hispanic/
White

R^2 :
0.4086
Adjusted:
0.3929

Variable	Estimate	Std. Err.	t-val.	p-val.	Boot Estimate	Boot Std. Err	Lower CI	Upper CI
Overall Food Insecurity Rate	0.490	0.091	5.372	2.19e-07	0.493	0.091	0.322	0.674
Cost per Meal	3.072	0.555	5.537	9.77e-08	3.08	0.579	1.97	4.17
SNAP Threshold	0.021	0.008	2.664	0.008	0.021	0.007	0.008	0.036
Overall Food Insecurity Rate	0.173	0.087	1.992	0.047	0.168	0.083	<u>-0.002</u>	0.320
Cost per Meal	1.918	0.520	3.689	0.0003	1.89	0.462	0.992	2.79
SNAP Threshold	0.034	0.007	4.618	6.50e-06	0.034	0.007	0.02	0.05



IMPORTANT FINDINGS/INTERPRETATIONS

Black/White:

- ❖ Cost per Meal is most significant (every dollar more causes a 3% gap increase)
- ❖ Overall FI rate also significant, (every % increase causes around a 0.5% gap incr.)
- ❖ SNAP Threshold slightly significant (every % higher causes 0.02% gap incr.)

Hispanic/White:

- ❖ Cost per Meal is still significant (every dollar more causes a 2% gap increase)
- ❖ Overall FI rate is least significant, (every % increase causes around a 0.2% gap incr.)
 - Only variable with a CI Interval that contains 0, indicating little significance
- ❖ SNAP Threshold slightly significant (every % higher causes 0.03% gap incr.)

In plain English:

- As meal cost (grocery store prices, etc) goes up, so does the gap in food insecurity rates. As overall food insecurity rate increases (the rate of all individuals, regardless of race), the gap widens.

For SNAP thresholds, our results (a positive coefficient) actually indicate that an increase in the threshold (meaning more people qualify) actually **INCREASES** the gap. This can be because of unequal SNAP access, paperwork/administrative barriers, structural barriers, immigrant status making them unable to access despite eligibility, and more.

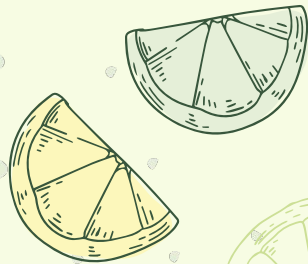


IS OUR MODEL **TRUSTWORTHY?**

- Our bootstrapping results are very similar to our actual regression results, meaning it is a trustworthy model.
- Our R^2 value indicates that around 40% of the variation in the FI gaps can be explained by our model. This is a reasonably strong fit for topics within the social science realm.
 - Data analysts online agree that with social sciences, there is high variability in variables



RESEARCH QUESTION CONCLUSIONS



1. Is it just that states with high overall FI have bigger gaps?
 - Yes, it is a significant predictor of the gaps, and a strong one for the BW gap.
2. Do economic factors (meal cost) matter more than policy factors (SNAP)?
 - Higher meal costs are strongly associated with bigger gaps.
 - SNAP threshold likely is increased based on a higher need and higher inequality.
 - So economic factors do matter more than policy, but both factors are still correlated to FI.
3. Are the main reasons for the Black-White gap the same as the Hispanic-White gap?
 - Generally, among the three that we studied, yes! Although to different extents, and magnitudes differ. The BW gap is more tied to the FI rate than the HW, and the opposite is true for the SNAP threshold.
 - Meal cost has a bigger impact on the BW gap than HW.

DATASET **LIMITATIONS**

- **All values are modeled estimates**, derived from survey and economic data, rather than direct counts.
- **Demographic breakdowns can be limited**, especially in smaller populations where sample sizes are too small for precise estimates.
- **Year-to-year comparisons may be influenced by shifts in economic conditions**, local economic conditions can influence both food insecurity and the model's estimates.
- **No population data provided**, if we want to make proper regional conclusions, this data will be necessary.



THANKS! Questions?



CREDITS: This presentation template was created by **Slidesgo**, and includes icons by **Flaticon**, and infographics & images by **Freepik**

WORK CITED

Ribar, D.C., Harris, V., Dewey, A., Dawes, S., and Engelhard, E. (2025). Map the Meal Gap: An Analysis of Local Food Insecurity and Food Costs in the United States in 2023. Feeding America National Organization.

Dewey, A., Harris, V., Hake, M., & Engelhard, E. (2024). Map the Meal Gap 2024: An Analysis of County and Congressional District Food Insecurity and County Food Cost in the United States in 2022. Feeding America.

Gundersen C., Ziliak, J. (2023). Hunger Among Adults Age 50-59 in 2021: An Annual Report. Report for Feeding America. Available from Feeding America:
<https://www.feedingamerica.org/research/state-senior-hunger>

Ziliak, J., Gundersen, C. (2023). The State of Senior Hunger in America in 2021: An Annual Report. Report for Feeding America. Available from Feeding America:
<https://www.feedingamerica.org/research/state-senior-hunger>

Gundersen, C., Strayer, M., Dewey, A., Hake, M., & Engelhard, E. (2023). Map the Meal Gap 2023: An Analysis of County and Congressional District Food Insecurity and County Food Cost in the United States in 2021. Feeding America.

Gundersen, C., Strayer, M., Dewey, A., Hake, M., & Engelhard, E. (2022). Map the Meal Gap 2022: An Analysis of County and Congressional District Food Insecurity and County Food Cost in the United States in 2020. Feeding America.

Gundersen, C., Strayer, M., Dewey, A., Hake, M., & Engelhard, E. (2021). Map the Meal Gap 2021: An Analysis of County and Congressional District Food Insecurity and County Food Cost in the United States in 2019. Feeding America.

Fadul, Mohammed Mustafa. (2022). Re: What is an acceptable R squared value for social sciences? . Retrieved from:
https://www.researchgate.net/post/What_is_an_acceptable_R_squared_value_for_social_science/s/62ff1b87709342352c0aef9b/citation/download.

